
SKILLS

- **Backend/Product:** Python, TypeScript, JavaScript, React, Node.js, REST APIs, GraphQL, microservices, web apps
- **Data Products:** data science tools, ML frameworks, dashboards, APIs, internal platforms, reusable libraries
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Snowflake, schema design, query optimization
- **Large Datasets:** Kafka, Airflow, ETL/ELT, batch/stream processing, validation, lineage, analytics pipelines
- **Cloud/Infra:** AWS, S3, Lambda, EKS, Docker, Kubernetes, Terraform, CI/CD, Linux, GitHub Actions
- **Distributed Systems:** highly available services, event-driven workflows, caching, parallel processing, API gateways
- **Quality/Ops:** testing strategy, code reviews, design patterns, OpenTelemetry, ELK, CloudWatch, SLOs, alerting
- **Collaboration:** requirements gathering, stakeholder documentation, technical specs, ADRs, demos, ownership

EXPERIENCE

- **Royal Bank of Canada (RBC)** Toronto, Canada
Staff Software Developer & Enterprise Architect – Data Products / Platform *Nov 2022 – Present*
 - **Data-Driven Product Platform:** Led 0-to-1 delivery of patent-pending enterprise blueprint platform using Node.js, TypeScript, React, REST APIs, real-time data rendering, rules-based recommendations, auto-certification workflows, and secure service matching for technical stakeholders.
 - **High-Scale Backend Data Services:** Architected telemetry platform using OpenTelemetry, Vector, Fluent Bit, Kafka, and Logstash ingesting 150 TB/day across 10,000+ systems; standardised schemas, routing, validation, alerting, and replayable ingestion for downstream analytics.
 - **Reusable Frameworks & Services:** Built shared platform abstractions, onboarding guides, schema contracts, and reusable service patterns adopted by 20+ teams; improved quality, maintainability, throughput, and developer experience across data producers and consumers.
 - **Stakeholder Requirements & Documentation:** Partner with architecture, product, infrastructure, security, and risk teams to understand challenges, gather requirements, model data, produce technical specs, demo iterations, and drive product improvements after launch.
 - **Ownership, Support & Reliability:** Own production readiness through CI/CD, monitoring, runbooks, incident triage, escalation paths, SLOs, data-readiness checks, code reviews, testing expectations, and continued support for regulated systems.
- **HealthCard (Acquired Startup)** Remote
Blockchain Architect & First Full Stack Engineer *Aug 2021 – Jul 2022*
 - **End-to-End Full-Stack Application:** Joined as first engineer and built React/TypeScript frontend, Node.js/Express APIs, AWS EKS deployment, audit logging, verification workflows, and trust-score data models for insurance validation and claims workflows.
 - **ML/Data Product Integration:** Integrated transformer-based document classification and monitoring for medical/insurance records, reducing manual review by 60%; built production APIs, feedback loops, and HIPAA-aligned audit trails for credential verification.
- **Canada Revenue Agency (CRA)** Ottawa, Canada (Remote)
Software Developer Co-op *Apr 2022 – Nov 2022*
 - **Production Web Modernization:** Migrated federal tax platform UI from Apache Struts to React/TypeScript and contributed to backend APIs serving millions of Canadians with accessibility, quality gates, secure delivery, and maintainable code standards.
- **Dalhousie University** Halifax, Canada
Research & Teaching Assistant *Jan 2021 – Aug 2022*
 - **Algorithms, Systems & Cloud:** Researched call-stack decision algorithms and Python tracing tools for compiler/runtime analysis; TA for Algorithms, Software Development, and graduate Advanced Cloud Computing.

PROJECTS

- **Data Science Workflow Platform:** Prototype Python/React platform for dataset ingestion, validation, feature generation, experiment outputs, and reusable dashboards for research-style analytics workflows.
- **Market Data Research Pipeline:** Python framework for ingesting market data, validating schemas, constructing query-friendly datasets, running backtests, and producing reproducible analytics for strategy research.
- **Portfolio Risk Optimization Framework:** Python research system for market data ingestion, portfolio allocation, Sharpe analysis, Monte Carlo simulation, risk-adjusted signals, and repeatable quantitative experiments.

EDUCATION

- **Dalhousie University** Halifax, Canada
MASc Applied Computer Science, GPA: 3.9/4.0 | TA: Algorithms, Advanced Cloud Computing *Jan 2021 – Aug 2022*
- **University of Toronto** Toronto, Canada
Enterprise Architecture Certification (3-semester program) *Completed 2024*
- **Gujarat Technological University** Ahmedabad, India
B.E. Computer Engineering, GPA: 3.92/4.0 *Aug 2016 – Aug 2020*